

Design for Earthquakes

In accordance with the Ontario Building Code 2012 (OBC), foundations must be designed to withstand a minimum earthquake force. Based on the measured average in-situ shear wave velocity of 794 m/s, the Site Classification for Seismic Site Response would be C (unless base of foundations or top of pile caps are located within 3 m of bedrock surface in which case Class B could be considered, however this is not likely practicable at this Site).

6.3.4 Liquefaction

Based on the encountered subsurface conditions and in-situ and laboratory testing conducted, summarized in sections above, the Site soils would not be considered liquefiable.

6.3.5 Foundation Dewatering

Generally, provided surficial slabs are located 0.2 m above existing site grades, we would not consider under slab drainage a necessity. However, given the close proximity of the observed static groundwater levels at this Site, we would recommend consideration of an underslab drainage network. Perimeter drainage around the structure and any below grade levels should be provided, tied to a frost-free outlet. If a basement is to be constructed, complete structural waterproofing such as a blindside system or a 'tanked' construction would likely be necessary. Surface grades should be sloped away from the building, and landscaped areas directly adjacent to the building should be capped with impermeable soils, if practicable.

6.1 Excavation and Temporary Shoring

Excavations up to a depth of approximately 1.5 to 2.5 m from the existing site grades may be necessary for construction of foundations, or for buried utilities. We would expect that sufficient area would be available to backslope excavations to this depth without the use of shoring at this Site. Once foundations are constructed, excavations should not extend within a zone extending 10 horizontal to 7 vertical from the bottom outside edge of these foundations, to prevent them from becoming undermined. Groundwater must be lowered to at least 0.5 m below the base of the lowest excavation to ensure a stable working surface.

All excavations and construction of any shoring should be carried out in accordance with the latest edition of the OHSA and Regulations for Construction Projects. The OHSA regulations require that if workers must enter an excavation deeper than 1.2 m, the excavation must be suitably sloped and/or braced in accordance with OHSA requirements. OHSA specifies the maximum slope of excavations into four broad soil types, summarized as follows:

Table 4 – OHSA Soil Types	Soil Type	Maximum Slope Inclination	Base of Slope Location
Type 1		1 horizontal to 1 vertical	Within 1.2 m of base of excavation
Type 2		1 horizontal to 1 vertical	Within 1.2 m of base of excavation
Type 3		1 horizontal to 1 vertical	From bottom of excavation
Type 4		3 horizontal to 1 vertical	From bottom of excavation

traffic loadings. The following parameters may be used in the design of temporary shoring:

Table 5 – Lateral Earth Pressure	Bulk Unit Weight (kN/m³)	Angle of Internal Friction (deg.)	Coefficient of Lateral Earth
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Parameters Soil			Pressure		
Type					
Ka			Kp		
Native Silty Clay	19.0	29	0.35	0.52	2.88
Native Silty Sand	21.0	32	0.31	0.47	3.25
Native Glacial Deposits (Till)	21.0	34	0.28	0.44	3.54
Compacted Gran. 'A'	21.0	34	0.28	0.44	3.54
Compacted Gran. 'B'	21.5	35	0.27	0.43	3.69
Type II					

LOADING

The following maximum loadings shall be considered for the design of the pile :

Axial load (Px) is 3 kips , Bending moment for the strong axis (Mx) is 6.5 kips*ft . Bending moment for the weak axis (My) is 0.4 kips*ft